Assignment 2: Git and Stream Temperature - July Analysis

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Data Loading

```
library(tidyverse)
library(lubridate)

# Load saved data

temp_data <- read.csv("data/temp_data.csv")

# Select and rename relevant columns
temp_data <- temp_data %>%
  select(
    agency_cd = monitoring_location_id,
    site_no = monitoring_location_id,
    Date = time,
    temp_C = value )

# Clean site names
temp_data$site_no <- gsub("USGS-", "", temp_data$site_no)

# Ensure proper format
temp_data$Date <- as.Date(temp_data$Date)
temp_data$month <- month(temp_data$Date, label = TRUE)
temp_data$year <- year(temp_data$Date)
```

Filter for July

```r
july_data <- temp_data %>%
  filter(month == "Jul")
```

Model: Temperature Trend Over Time

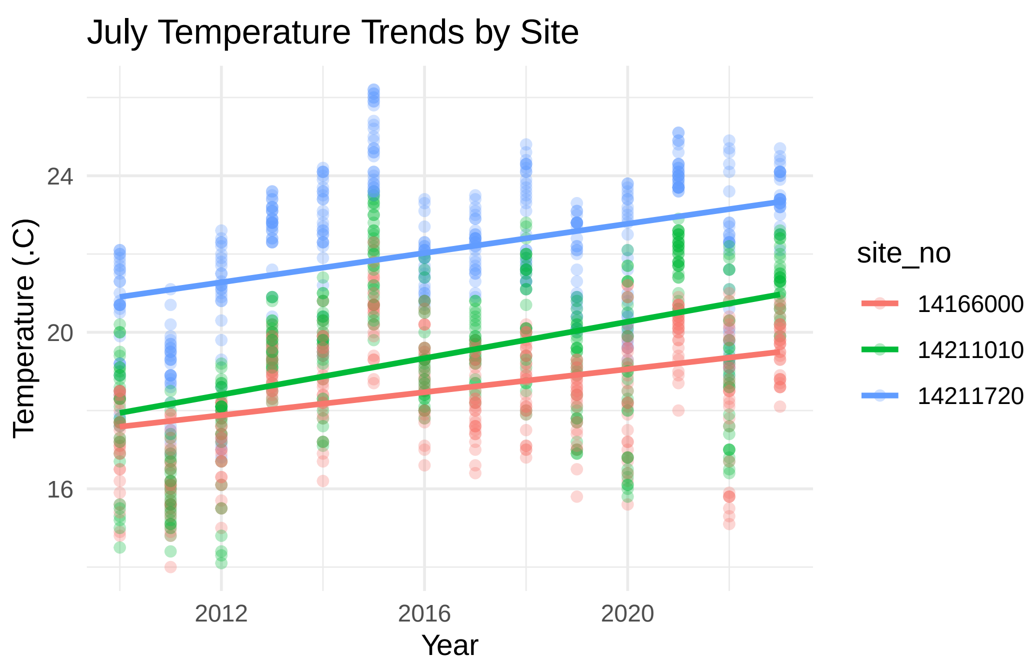
We use a generalized liner model (GLM) to estimate whether stream temperature is changing over time and whether trends differ across sites.

```r
july_model <- glm(temp_C ~ year * site_no,
                  data = july_data)

summary(july_model)
```

```

Call:
glm(formula = temp_C ~ year * site_no, data = july_data)

Coefficients:
                       Estimate Std. Error t value Pr(>|t|)    
(Intercept)          -278.16357   39.88799  -6.974 4.91e-12 ***
year                    0.14714    0.01978   7.439 1.85e-13 ***
site_no14211010      -172.34776   56.41460  -3.055  0.00230 ** 
site_no14211720       -76.71493   56.49555  -1.358  0.17473    
year:site_no14211010    0.08592    0.02798   3.071  0.00218 ** 
year:site_no14211720    0.03982    0.02802   1.421  0.15550    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 2.759493)

    Null deviance: 7333  on 1298  degrees of freedom
Residual deviance: 3568  on 1293  degrees of freedom
AIC: 5012.9

Number of Fisher Scoring iterations: 2
```

Visualization of Trends

```
ggplot(july_data, aes(x = year, y = temp_C,
                      color = site_no)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(
    title = "July Temperature Trends by Site",
    x = "Year",
    y = "Temperature (°C)" ) +
  theme_minimal()
```



Interpretation of Results

July stream temperatures show a clear increasing trend over time. The coefficient for year is positive and highly significant ($p < 0.001$), indicating that temperatures are rising at a rate of approximately 0.15 ° C per year at the baseline site (site 14166000).

There are also differences in warming rates between sites. Site 14211010 shows a significantly greater increase in temperature over time compared to the baseline site, with an additional warming rate of about 0.086 ° C per year. This suggests that this site is warming more rapidly than the others.

In contrast, site 14211720 shows a slightly higher warming rate than the baseline site, but this difference is not statistically significant. Therefore, there is no strong evidence that this site is warming at a different rate.

Overall, the results indicate that July stream temperatures are increasing over time, with some sites experiencing faster warming than others.